**PRACTICAL**

You are starting your PhD at a virology lab. From your conversation with the PI in the lab you learned that they are working on the discovery of protein reagents to target important proteins in viruses. In particular, the group is working toward the targeting of the hemagglutinin receptor (***HA***) of influenza (flu). The PI mentioned that they have designed a protein they call ***HD100*** that binds in a one-to-one ratio to HA with high specificity and affinity.

Your PI mention that they need structural information of the HA-HD100 complex to guide a further maturation of HD100 and inform potential changes to improve the binding affinity. You mention that during your MsC, you mastered the techniques of protein docking and that indeed docking could be a useful approach to derive structural models. You agree to take on this as a sort of side-project

During your conversation you learn that HD100 competes with the binding of a neutralizing antibody: Fa CR6261. The X-ray crystal structure of the complex HA-Fab CR6261 is known, as it was solved in the lab in the past. Your PI share with you the coordinates of these complex (**HA_Ab.pdb**). Further, you learn that the designed protein, HD100, was derived from a protein called APC36109 from *Bacillus stearothermophilus*. The function of this protein is unknown although the crystal structure of the protein is known: PDB code 1U84. They performed a number of mutations on 1U84 to obtain the binder of HA, i.e. HD100. The alignment between HD100 (designed – binds HA) and 1U84 (wildtype – doesn't bind HA) sequences as follow:

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The sequence alignment of HD100 and 1U84:

HD100    GQQLNRLLLEWIGAWDPFGLGKDAYDVEAEAVLQAVYETESAFDLAMRIMWIYVF
         GQQLNRLLLEWIGAWDPFGLGKDAYDVEA +VLQAVYETE A  LA RI  IY F
1U84     GQQLNRLLLEWIGAWDPFGLGKDAYDVEAASVLQAVYETEDARTLAARIQSIYEF

HD100    AFNRPIPFSHAQKLARRLLELKQAAS
         AF+ PIPF H  KLARRLLELKQAAS
1U84     AFDEPIPFPHCLKLARRLLELKQAAS
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**Files provided:**

1. Coordinates of HA bound to neutralizing antibody: HA_Ab.pdb
2. FastA file of HD100 sequence: HD100.fa
3. Coordinate of APC36109: 1U84.pdb

**Based on all this information you should provide structural models of the binary complex HD1000 and HA using both unbiased and biased docking. The tasks to be completed are:**

**A. Unbiased docking of HA and HD100.**

1. Obtain the coordinates of both HA (0.5pts) and HD100 (1.5 pts) needed to perform the docking. For the purpose of the docking: HA will be the receptor (chain id: 'A') and

HD100 will be the ligand (chain id: 'B'.) Call your files: HA.pdb and HD100.pdb respectively.

2. Perform an unbiased rigid body docking using any of the methods used during the practical sessions using the coordinates of receptor and ligand obtained in (1) (1.0 pts.)

3. Extract and save the top 10 docking poses according to the native scoring of your docking method of choice (0.5 pts.) **Note: given the size of HA receptor, the use of PatchDock without restraints is recommended.**

## B. Biased docking of HA and HD100 using PatchDock.

1. Based on the information provided above, identify both in the HA receptor and HD100 designed protein the residues potentially important for the interaction. Include this information in a text file called 'putative_interface.txt' (2 pts.).

2. Prepare the two restraints files needed by PatchDock from the residues identified in (1). Call these files s1.txt and s2.txt (1 pts.)

3. Prepare the parameter file needed by PatchDock using a value of 4.0 as clustering RMSD cut-off (0.5pts.) and perform the **biased** docking. (1.0 pts.)

4. Extract and save the top 10 docking poses (0.5)

It is important to document the process you followed to obtain the results in writing. You don't have to explain the methodology neither describe how you run the programs. Moreover, keep all the intermediate steps as well as the final results. This will help us understanding the process you followed and/or debug potential errors. Place the intermediary files in a directory called 'scratch' and the final results in a directory called 'results'

**The final results directory has to include the following:**

- A brief 'README.txt' files documenting the different steps you followed to accomplish the different tasks

- From the A exercise:
  - The coordinates of both HA and HD100 proteins in PDB format.
  - The file with the unbiased docking output. For instance, if you used ZDOCK, it should be the file containing the information of all docking poses (e.g. T26.zdock from practical SBI-17)
  - The Top 10 docking poses for the unbiased docking run.

- From the B exercise
  - The file with the putative interface: putative_interface.txt
  - The s1.txt and s2.txt restraints files used by PatchDock.
  - The params.txt and PatchDock output file.
  - The top 10 docking poses for the biased docking run.